

YOLOv8 Fine-Tuning on DoclayLaynet

Document layout analysis plays a crucial role in understanding the structure of complex documents by identifying elements such as titles, paragraphs, tables, figures, and lists. The DocLayNet dataset provides a large collection of real-world document images annotated for layout detection tasks, making it a valuable resource for training and evaluating deep learning models.

In this notebook, we focus on fine-tuning the YOLOv8 object detection model on a curated subset of the DocLayNet dataset, which has been uploaded to the Hugging Face Hub. Our goal is to train a lightweight yet powerful model capable of accurately detecting layout components in historical or scanned documents.

- We'll walk through:
- Setting up the dataset and configuration
- Preparing the YOLOv8 environment
- Training and validation
- Evaluating performance and visualizing results This fine-tuned model can be further used for downstream tasks like intelligent document processing, OCR post-processing, and structured data extraction.

Install Required libraries

```
!pip install datasets
from datasets import load_dataset
!pip install ultralytics
import numpy as np
import os
import cv2
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from ultralytics import YOLO
```

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Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.11/dist-packages (from jinja2->torch>=1.8.0-
>ultralitics) (3.0.2)

```

checking for GPU(was originally ran on collab)

```

!nvidia-smi

Wed Apr  9 14:17:16 2025
+-----+
+-----+
| NVIDIA-SMI 550.54.15                  Driver Version: 550.54.15
| CUDA Version: 12.4                    |
+-----+-----+
+-----+
| GPU Name                               Persistence-M | Bus-Id        Disp.A |
| Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap |      Memory-Usage |
| GPU-Util  Compute M. |
|                               |                      |
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|  0  Tesla T4                               Off |  00000000:00:04.0 Off |
0 |
| N/A    41C    P8              9W /   70W |      0MiB /  15360MiB |
0%      Default |
|                               |                      |
N/A |
+-----+-----+
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+-----+
| Processes:
|
| GPU   GI    CI          PID    Type    Process name
GPU Memory |
|      ID      ID
Usage      |

```

```
|
=====
=====|
| No running processes found
|
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```

Dataset Details

- Name: DocLayNet Subset
- Source: Uploaded to Hugging Face Hub
- Format: COCO-style JSON annotations
- Classes: ['text', 'title', 'list', 'table', 'figure', 'footer', 'header', ...]

```
ds = load_dataset("Epicquest97/DocLayNet3")
import os
import cv2

def convert_to_yolo_format(bbox, img_width, img_height):
    x, y, w, h = bbox
    x_center = (x + w / 2) / img_width
    y_center = (y + h / 2) / img_height
    w_norm = w / img_width
    h_norm = h / img_height
    return x_center, y_center, w_norm, h_norm

def save_yolo_labels(data_split, output_dir):
    os.makedirs(f"{output_dir}/images/{data_split}", exist_ok=True)
    os.makedirs(f"{output_dir}/labels/{data_split}", exist_ok=True)

    for i, sample in enumerate(ds[data_split]):
        img = sample['image']
        bboxes = sample['bboxes']
        category_ids = sample['category_id']

        # Convert to NumPy array if necessary
        if not isinstance(img, np.ndarray):
            img = np.array(img)

        img_path = f"{output_dir}/images/{data_split}/{i}.jpg"
        label_path = f"{output_dir}/labels/{data_split}/{i}.txt"
        cv2.imwrite(img_path, cv2.cvtColor(img, cv2.COLOR_RGB2BGR))

        with open(label_path, 'w') as f:
            for bbox, category_id in zip(bboxes, category_ids):
                x_center, y_center, w_norm, h_norm =
convert_to_yolo_format(bbox, img.shape[1], img.shape[0])
                f.write(f"{category_id} {x_center} {y_center} {w_norm}
{h_norm}\n")
```

```

save_yolo_labels('train', 'dataset')
save_yolo_labels('val', 'dataset')
save_yolo_labels('test', 'dataset')

/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(

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{"model_id": "d3401fbe109c4d08b6ea1eec4d536c0e", "version_major": 2, "version_minor": 0}

```

Dataset Visualisation

```

def plot_sample_with_labels(example, class_names):
    img = example['image']
    bboxes = example['bboxes']
    labels = example['category_id'] # or whatever your label field is called

    fig, ax = plt.subplots(1, 1, figsize=(10, 10))
    ax.imshow(img)

    for bbox, label in zip(bboxes, labels):
        x, y, w, h = bbox
        rect = patches.Rectangle((x, y), w, h, linewidth=1.5,

```

```
edgecolor='lime', facecolor='none')
    ax.add_patch(rect)
    ax.text(x, y - 5, class_names[label], color='lime',
fontsize=9, backgroundcolor='black')

    ax.axis('off')
    plt.tight_layout()
    plt.show()

# Example usage:
class_names = ['text', 'title', 'list', 'table', 'figure',
'header', '1', '2', '3', '4', '5']
for i in range(10, 13):
    plot_sample_with_labels(ds['train'][i], class_names)
```

5

4

5

1

5

4

1

3

3

1

1

5 flaw in (2) may be. Not in the signs, but rather, in the lack of a term. A term that shouldn't be a constant, but dependent of n . A term that added up to the other two would not only make N_3 positive for $n \gg 1$, but actually equal to six. So there must be an additional source of degrees of freedom which Einstein missed to consider. If we could identify where this source of degrees of freedom was, we would have our problem solved.

5 A group of n particles may rotate in space without dissatisfying the condition that the distances between the particles remain constant. However, it is meaningless to talk about rotations without first establishing an adequate reference frame. To do so we arbitrarily selected three particles from S_3 ; the points were the centers of mass of these particles are located generate a plane P in three-dimensional space. And the vector $\mathbf{v}$, which is orthogonal to P , designates an arbitrary direction in space. We must point out that we are defining $\mathbf{v}$ as a fixed vector, and that it is perpendicular to P in its original position, but as S_3 rotates, this perpendicularity relation will be lost. Therefore, it is convenient to make a copy of P , which we will call P' , and hold this copy fixed in the original position of P . Thus $\mathbf{v}$ will always be orthogonal to P' . By considering the plane P' and its normal vector, we are defining a three-dimensional coordinate system.

5 Now, if we choose two particles, different from the ones used to generate the plane, the line that joins their centers of mass is a possible rotation axis for S_3 . And since the number of ways in which pairs may be chosen from a set of $n - 3$ particles is

table

$$C_{n-3}^2 = \frac{(n-3)!}{2!(n-5)!} = \frac{(n-3)(n-4)}{2}, \quad (6)$$

for $n \geq 3$.

5 There will be an equal number of such axes. Each of this axes forms with the direction of the vector $\mathbf{v}$ an angle φ_i which is a function of time and determines a possible rotation of the system. In general, the different φ_i will not hold relations of linear dependence.

5 We believe that the number of φ_i allowed to S_3 for a given value of n is the term missing in Einstein's calculation, and we propose that the number of degrees of freedom for the system S_3 is given by:

table

$$N_3 = 3n - \frac{n(n-1)}{2} + \frac{(n-3)(n-4)}{2} = 6, \quad (7)$$

when $n \geq 3$.

5 However, (2) seems to be the correct expression for $n = 2$. It also works for $n = 3$ and $n = 4$, which is not surprising, since for this value of n the last term in expression (7) is equal to zero, so (7) and (2) are equivalent.

5 Once we had developed this expressions, we were curious on whether, by following the

5 Our stockholders' equity has been reduced by \$11.2 billion from the annual year-end measurement of the funded status of our postretirement benefit plans, inclusive of the December 31, 2011 adjustment of \$2.9 billion. These noncash, after-tax amounts primarily represent net actuarial losses resulting from declines in discount rates and differences between actual experience and our actuarial assumptions, which will be amortized to expense in future periods. During 2011, \$666 million of these amounts was recognized as a component of our postretirement benefit plans expense and \$812 million is expected to be recognized as expense in 2012.

5 We expect that our 2012 pension expense will increase to \$1.9 billion as compared with 2011 pension expense of \$1.8 billion, primarily due to an increase in the amortization of net actuarial gains and losses caused by the decrease in the discount rate mentioned above.

5 The discount rate assumption we select at the end of each year is based on our best estimates and judgment. A reasonably possible change of plus or minus 25 basis points in the 4.75% discount rate assumption at December 31, 2011, with all other assumptions held constant, would have decreased or increased the amount of the qualified pension benefit obligation we recorded at the end of 2011 by approximately \$1.3 billion, which would have resulted in an after-tax increase or decrease in stockholders' equity at the end of the year of approximately \$850 million. If the 4.75% discount rate at December 31, 2011 that was used to compute the expected 2012 expense for our qualified defined benefit pension plans had been 25 basis points higher or lower, with all other assumptions held constant, the amount of expense projected for 2012 would be lower or higher by approximately \$125 million.

3 Funding Considerations

5 The PPA became applicable to us and other large U.S. defense contractors beginning in 2011 and had the effect of accelerating the required amount of annual pension plan contributions. We made contributions related to our qualified defined benefit pension plans of \$2.3 billion in 2011, \$2.2 billion in 2010, and \$1.5 billion in 2009. We recovered \$899 million in 2011, \$988 million in 2010 and \$580 million in 2009 as CAS costs. Amounts funded under CAS are recovered over time through the pricing of our products and services on U.S. Government contracts, including FMS, and are recognized in our cost of sales and net sales. Amounts contributed in excess of the CAS funding requirements, over \$3.0 billion, are considered to be prepayment credits under the CAS rules.

5 We expect to make contributions of \$1.1 billion related to our qualified defined benefit pension plans in 2012 and anticipate recovering \$1.1 billion as CAS cost in 2012 which is consistent with our anticipated contributions. We may review options for further contributions in 2012.

5 The CAS Board published its revised pension accounting rules (CAS Harmonization) with an effective date of February 27, 2012 to better align the recovery of pension contributions, including prepayment credits, on U.S. Government contracts with the accelerated funding requirements of the PPA. The CAS Harmonization rules will increase our CAS cost beginning in 2013. There is a transition period during which the cost impact of the new rules will be phased in, with the full impact occurring in 2017. While we expect our 2013 CAS costs to be higher than our estimate for 2012 of \$1.1 billion, the estimated incremental impact of CAS Harmonization in 2013 will be a very modest cost increase, with much larger increases occurring successively in years 2014 through 2017.

5 Based upon current assumptions which may change, the increase in CAS costs caused by CAS Harmonization should result in increased earnings a few years from now, as our CAS costs should be in excess of the pension expense we record under GAAP. Accordingly, our non-cash FAS/CAS pension adjustment, discussed further in the "Discussion of Business Segments" section above, should eventually increase earnings rather than decrease earnings as it has the past few years. In addition, the increase in CAS costs should eventually cause our CAS costs to be greater than our pension contributions as we recover the prepayment credits, which should increase our cash flow from operations.

3 Environmental Matters

5 We are a party to various agreements, proceedings, and potential proceedings for environmental cleanup issues, including matters at various sites where we have been designated a potentially responsible party (PRP) by the EPA or by a state agency. At the end of 2011, the total amount of liabilities recorded on our Balance Sheet for environmental matters was \$932 million. We have recorded receivables totaling \$808 million at December 31, 2011 for the portion of environmental costs that are probable of future recovery in pricing of our products and services for agencies of the U.S. Government, as discussed below. The amount that is expected to be allocated to our non-U.S. Government contracts or that is determined to be unallowable for pricing under U.S. Government contracts has been expensed through cost of sales. We project costs and recovery of costs over approximately twenty years.

header

CAS

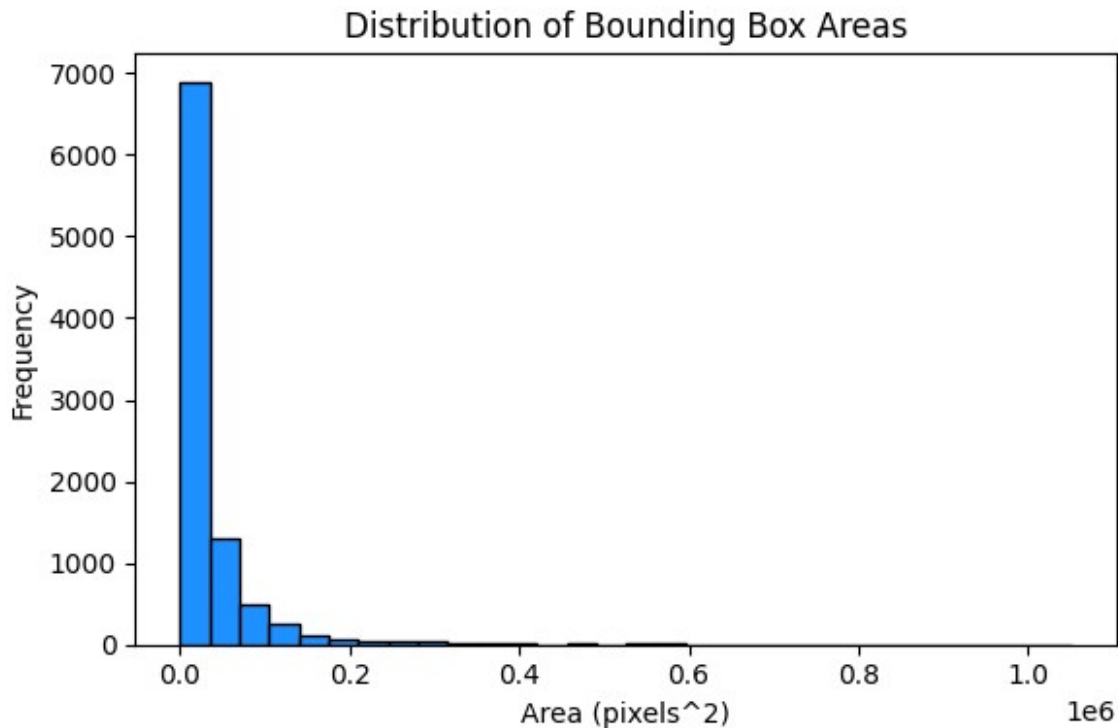
```
import numpy as np

box_sizes = []

for ex in ds['train']:
    for box in ex['bboxes']:
        _, _, w, h = box
        box_sizes.append(w * h)

plt.figure(figsize=(6,4))
plt.hist(box_sizes, bins=30, color='dodgerblue', edgecolor='black')
plt.title("Distribution of Bounding Box Areas")
```

```
plt.xlabel("Area (pixels^2)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



Running the YOLO on Costum Dataset

```
!yolo task=detect mode=train data=dataset/data.yaml model=yolov8s.pt
epochs=50 imgsz=640 batch=48 save=True
```

Ultralytics 8.3.105 Python-3.11.11 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

```
engine/trainer: task=detect, mode=train, model=yolov8s.pt,
data=dataset/data.yaml, epochs=50, time=None, patience=100, batch=48,
imgsz=640, save=True, save_period=-1, cache=False, device=None,
workers=8, project=None, name=train3, exist_ok=False, pretrained=True,
optimizer=auto, verbose=True, seed=0, deterministic=True,
single_cls=False, rect=False, cos_lr=False, close_mosaic=10,
resume=False, amp=True, fraction=1.0, profile=False, freeze=None,
multi_scale=False, overlap_mask=True, mask_ratio=4, dropout=0.0,
val=True, split=val, save_json=False, conf=None, iou=0.7, max_det=300,
half=False, dnn=False, plots=True, source=None, vid_stride=1,
stream_buffer=False, visualize=False, augment=False,
agnostic_nms=False, classes=None, retina_masks=False, embed=None,
show=False, save_frames=False, save_txt=False, save_conf=False,
save_crop=False, show_labels=True, show_conf=True, show_boxes=True,
line_width=None, format=torchscript, keras=False, optimize=False,
```

```

int8=False, dynamic=False, simplify=True, opset=None, workspace=None,
nms=False, lr0=0.01, lrf=0.01, momentum=0.937, weight_decay=0.0005,
warmup_epochs=3.0, warmup_momentum=0.8, warmup_bias_lr=0.1, box=7.5,
cls=0.5, dfl=1.5, pose=12.0, kobj=1.0, nbs=64, hsv_h=0.015, hsv_s=0.7,
hsv_v=0.4, degrees=0.0, translate=0.1, scale=0.5, shear=0.0,
perspective=0.0, flipud=0.0, fliplr=0.5, bgr=0.0, mosaic=1.0,
mixup=0.0, copy_paste=0.0, copy_paste_mode=flip,
auto_augment=randaugument, erasing=0.4, crop_fraction=1.0, cfg=None,
tracker=botsort.yaml, save_dir=runs/detect/train3
Downloading https://ultralytics.com/assets/Arial.ttf to
'/root/.config/Ultralytics/Arial.ttf'...
100% 755k/755k [00:00<00:00, 21.4MB/s]
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
E0000 00:00:1744208495.281356      2323 cuda_dnn.cc:8310] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1744208495.354186      2323 cuda_blas.cc:1418] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered
Overriding model.yaml nc=80 with nc=12

```

	from	n	params	module
arguments				
0	-1	1	928	ultralytics.nn.modules.conv.Conv
[3, 32, 3, 2]				
1	-1	1	18560	ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]				
2	-1	1	29056	ultralytics.nn.modules.block.C2f
[64, 64, 1, True]				
3	-1	1	73984	ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]				
4	-1	2	197632	ultralytics.nn.modules.block.C2f
[128, 128, 2, True]				
5	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
6	-1	2	788480	ultralytics.nn.modules.block.C2f
[256, 256, 2, True]				
7	-1	1	1180672	ultralytics.nn.modules.conv.Conv
[256, 512, 3, 2]				
8	-1	1	1838080	ultralytics.nn.modules.block.C2f
[512, 512, 1, True]				
9	-1	1	656896	
ultralytics.nn.modules.block.SPPF				[512, 512, 5]
10	-1	1	0	
torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
11	[-1, 6]	1	0	
ultralytics.nn.modules.conv.Concat				[1]

```

12          -1  1    591360  ultralytics.nn.modules.block.C2f
[768, 256, 1]
13          -1  1         0
torch.nn.modules.upsampling.Upsample      [None, 2, 'nearest']

14          [-1, 4]  1         0
ultralytics.nn.modules.conv.Concat        [1]

15          -1  1    148224  ultralytics.nn.modules.block.C2f
[384, 128, 1]
16          -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
17          [-1, 12]  1         0
ultralytics.nn.modules.conv.Concat        [1]

18          -1  1    493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
19          -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
20          [-1, 9]  1         0
ultralytics.nn.modules.conv.Concat        [1]

21          -1  1    1969152  ultralytics.nn.modules.block.C2f
[768, 512, 1]
22          [15, 18, 21]  1    2120692
ultralytics.nn.modules.head.Detect        [12, [128, 256, 512]]

```

Model summary: 129 layers, 11,140,244 parameters, 11,140,228 gradients, 28.7 GFLOPs

Transferred 349/355 items from pretrained weights

TensorBoard: Start with 'tensorboard --logdir runs/detect/train3',
view at <http://localhost:6006/>

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading

<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolol11n.pt>
to 'yolol11n.pt'...

100% 5.35M/5.35M [00:00<00:00, 90.1MB/s]

AMP: checks passed ☐

train: Scanning /content/dataset/labels/train... 693 images, 3
backgrounds, 2 corrupt: 100% 693/693 [00:00<00:00, 2219.82it/s]
train: WARNING Δ /content/dataset/images/train/272.jpg: ignoring
corrupt image/label: non-normalized or out of bounds coordinates [1.0006 1.0006]

train: WARNING Δ /content/dataset/images/train/556.jpg: ignoring
corrupt image/label: non-normalized or out of bounds coordinates [1.0002]

train: New cache created: /content/dataset/labels/train.cache

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
 val: Scanning /content/dataset/labels/val... 64 images, 0 backgrounds, 0 corrupt: 100% 64/64 [00:00<00:00, 1664.89it/s]
 val: New cache created: /content/dataset/labels/val.cache
 Plotting labels to runs/detect/train3/labels.jpg...
 optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...
 optimizer: AdamW(lr=0.000625, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.000375), 63 bias(decay=0.0)
 TensorBoard: model graph visualization added 
 Image sizes 640 train, 640 val
 Using 2 dataloader workers
 Logging results to runs/detect/train3
 Starting training for 50 epochs...

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	1/50	10.9G	2.817	4.858	2.127	469
640:	100% 15/15	[00:14<00:00, 1.00it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:06<00:00, 6.45s/it]		
		all	64	1105	0.247	0.0378
0.0277	0.0216					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	2/50	10.9G	1.639	2.214	1.325	516
640:	100% 15/15	[00:12<00:00, 1.20it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:03<00:00, 3.81s/it]		
		all	64	1105	0.476	0.221
0.157	0.0865					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	3/50	11G	1.332	1.679	1.189	513
640:	100% 15/15	[00:12<00:00, 1.24it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:02<00:00, 2.53s/it]		
		all	64	1105	0.294	0.369
0.292	0.17					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	4/50	11G	1.265	1.524	1.156	445
640:	100% 15/15	[00:12<00:00, 1.23it/s]				

		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.42s/it]				
		all	64	1105	0.308	0.333
0.272	0.155					
	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	5/50	11G	1.196	1.421	1.129	328
640:	100% 15/15	[00:11<00:00, 1.25it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.67s/it]				
		all	64	1105	0.67	0.291
0.315	0.168					
	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	6/50	11.1G	1.165	1.363	1.107	460
640:	100% 15/15	[00:11<00:00, 1.27it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.12s/it]				
		all	64	1105	0.414	0.334
0.272	0.148					
	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	7/50	11.1G	1.172	1.356	1.105	398
640:	100% 15/15	[00:12<00:00, 1.21it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.19s/it]				
		all	64	1105	0.373	0.254
0.225	0.119					
	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	8/50	11.1G	1.118	1.301	1.075	568
640:	100% 15/15	[00:12<00:00, 1.22it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.16s/it]				
		all	64	1105	0.533	0.33
0.325	0.184					
	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	9/50	11.2G	1.092	1.25	1.06	372
640:	100% 15/15	[00:11<00:00, 1.25it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1 [00:01<00:00, 1.02s/it]				
		all	64	1105	0.489	0.45
0.425	0.232					

Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	10/50	11.2G	1.094	1.263	1.065	306
100%	15/15	[00:12<00:00, 1.21it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.08s/it]		
		all	64	1105	0.585	0.469
0.471	0.269					
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	11/50	11.3G	1.012	1.161	1.031	470
100%	15/15	[00:12<00:00, 1.21it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.26s/it]		
		all	64	1105	0.536	0.504
0.493	0.293					
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	12/50	11.3G	1.002	1.152	1.028	442
100%	15/15	[00:12<00:00, 1.24it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.31s/it]		
		all	64	1105	0.606	0.471
0.492	0.279					
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	13/50	11.3G	1.031	1.165	1.028	475
100%	15/15	[00:12<00:00, 1.19it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.15s/it]		
		all	64	1105	0.609	0.505
0.531	0.328					
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	14/50	11.4G	0.9573	1.086	1.007	448
100%	15/15	[00:12<00:00, 1.21it/s]				
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:00<00:00, 1.05it/s]		
		all	64	1105	0.559	0.492
0.498	0.313					
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	15/50	12G	0.9524	1.066	1.004	390
100%	15/15	[00:12<00:00, 1.19it/s]				
		Class	Images	Instances	Box(P	R

mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.02s/it]			
0.538	0.308	all	64	1105	0.578	0.513

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	16/50	12G	0.9507	1.025	0.9964
640:	100%	15/15	[00:12<00:00, 1.21it/s]		527
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.05s/it]		
0.507	0.294	all	64	1105	0.568
					0.498

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	17/50	12.1G	0.9057	0.9954	0.9813
640:	100%	15/15	[00:12<00:00, 1.20it/s]		412
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.02s/it]		
0.528	0.319	all	64	1105	0.65
					0.465

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	18/50	12.1G	0.9056	1.003	0.9878
640:	100%	15/15	[00:12<00:00, 1.25it/s]		485
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.44s/it]		
0.551	0.315	all	64	1105	0.651
					0.515

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	19/50	12.1G	0.8881	0.9701	0.9749
640:	100%	15/15	[00:12<00:00, 1.24it/s]		396
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.08s/it]		
0.576	0.361	all	64	1105	0.558
					0.558

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	20/50	12.2G	0.8879	0.974	0.9729
640:	100%	15/15	[00:12<00:00, 1.17it/s]		408
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.11s/it]		
0.591	0.36	all	64	1105	0.643
					0.566

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
-------	---------	----------	----------	-----------	-----------

Size	21/50	12.2G	0.8769	0.9449	0.972	386
640:	100%	15/15	[00:12<00:00, 1.23it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.09s/it]		
		all	64	1105	0.572	0.573
0.58	0.37					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	22/50	12.2G	0.8414	0.9021	0.9618	484
640:	100%	15/15	[00:12<00:00, 1.24it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.04s/it]		
		all	64	1105	0.642	0.546
0.562	0.348					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	23/50	12.3G	0.8657	0.9193	0.9624	501
640:	100%	15/15	[00:12<00:00, 1.20it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.02s/it]		
		all	64	1105	0.668	0.51
0.564	0.359					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	24/50	12.3G	0.8337	0.9206	0.9578	397
640:	100%	15/15	[00:12<00:00, 1.20it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.02s/it]		
		all	64	1105	0.64	0.523
0.57	0.368					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	25/50	12.3G	0.8225	0.8772	0.9537	441
640:	100%	15/15	[00:12<00:00, 1.25it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.41s/it]		
		all	64	1105	0.675	0.563
0.604	0.4					

	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size	26/50	12.4G	0.809	0.8661	0.9509	362
640:	100%	15/15	[00:12<00:00, 1.24it/s]			
		Class	Images	Instances	Box(P	R
mAP50	mAP50-95):	100%	1/1	[00:01<00:00, 1.18s/it]		

0.598	0.381	all	64	1105	0.559	0.589
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	27/50	12.4G	0.7979	0.8475	0.9481	415
100%	15/15	[00:12<00:00, 1.23it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95):	100%	1/1	[00:01<00:00, 1.14s/it]			
0.664	0.425	all	64	1105	0.672	0.637
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	28/50	12.5G	0.7972	0.8501	0.9511	302
100%	15/15	[00:12<00:00, 1.21it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95):	100%	1/1	[00:00<00:00, 1.12it/s]			
0.643	0.411	all	64	1105	0.589	0.599
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	29/50	12.5G	0.81	0.8279	0.948	466
100%	15/15	[00:12<00:00, 1.21it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95):	100%	1/1	[00:00<00:00, 1.04it/s]			
0.64	0.428	all	64	1105	0.662	0.545
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	30/50	12.5G	0.7987	0.8512	0.9483	356
100%	15/15	[00:12<00:00, 1.19it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95):	100%	1/1	[00:00<00:00, 1.04it/s]			
0.61	0.365	all	64	1105	0.577	0.582
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
640:	31/50	12.6G	0.7727	0.8077	0.937	440
100%	15/15	[00:11<00:00, 1.25it/s]				
mAP50	Class	Images	Instances	Box(P	R	
mAP50-95):	100%	1/1	[00:01<00:00, 1.11s/it]			
0.636	0.424	all	64	1105	0.694	0.563
Size	Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances

32/50	12.6G	0.7626	0.7939	0.938	392
640: 100%	15/15	[00:12<00:00, 1.21it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 1.03it/s]		
	all	64	1105	0.609	0.645
0.639	0.416				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
33/50	12.6G	0.768	0.7961	0.9323	399
640: 100%	15/15	[00:12<00:00, 1.23it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.39s/it]		
	all	64	1105	0.669	0.583
0.621	0.394				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
34/50	12.7G	0.7543	0.7727	0.9337	504
640: 100%	15/15	[00:12<00:00, 1.19it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.03s/it]		
	all	64	1105	0.71	0.564
0.646	0.421				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
35/50	12.7G	0.7237	0.7466	0.9273	376
640: 100%	15/15	[00:12<00:00, 1.19it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:00<00:00, 1.15it/s]		
	all	64	1105	0.698	0.62
0.69	0.443				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
36/50	12.7G	0.7542	0.7585	0.931	504
640: 100%	15/15	[00:13<00:00, 1.15it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.68s/it]		
	all	64	1105	0.704	0.588
0.67	0.437				

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
37/50	12.8G	0.7345	0.7237	0.9211	400
640: 100%	15/15	[00:14<00:00, 1.07it/s]			
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	1/1	[00:01<00:00, 1.03s/it]		
	all	64	1105	0.703	0.632

```

0.687      0.455
Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
38/50      13.4G      0.7197      0.7221      0.9261      486
640: 100% 15/15 [00:14<00:00, 1.06it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.21s/it]
all        64        1105      0.692      0.572
0.641      0.415

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
39/50      11.6G      0.7176      0.7437      0.9243      464
640: 100% 15/15 [00:13<00:00, 1.14it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:00<00:00, 1.01it/s]
all        64        1105      0.678      0.626
0.67      0.424

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
40/50      12.3G      0.6926      0.7077      0.915      476
640: 100% 15/15 [00:14<00:00, 1.03it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.20s/it]
all        64        1105      0.694      0.584
0.66      0.426

```

```

Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3,
method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
41/50      12.3G      0.7592      0.7242      0.9199      208
640: 100% 15/15 [00:19<00:00, 1.28s/it]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.68s/it]
all        64        1105      0.71      0.594
0.668      0.448

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
42/50      12.3G      0.7154      0.6931      0.9075      240
640: 100% 15/15 [00:13<00:00, 1.08it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.05s/it]
all        64        1105      0.674      0.648

```

```

0.679      0.466
Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
43/50      12.3G      0.6952      0.6554      0.8995      255
640: 100% 15/15 [00:13<00:00, 1.08it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.10s/it]
all        64        1105      0.598      0.644
0.649      0.428

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
44/50      12.3G      0.6962      0.6516      0.8978      256
640: 100% 15/15 [00:13<00:00, 1.12it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.29s/it]
all        64        1105      0.643      0.628
0.667      0.434

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
45/50      12.3G      0.6779      0.6403      0.8924      263
640: 100% 15/15 [00:14<00:00, 1.05it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.21s/it]
all        64        1105      0.731      0.612
0.698      0.47

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
46/50      12.3G      0.6595      0.6182      0.8921      202
640: 100% 15/15 [00:13<00:00, 1.13it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.09s/it]
all        64        1105      0.73      0.644
0.713      0.492

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
47/50      12.3G      0.6615      0.6033      0.8866      258
640: 100% 15/15 [00:12<00:00, 1.16it/s]
Class      Images  Instances  Box(P      R
mAP50      mAP50-95): 100% 1/1 [00:01<00:00, 1.61s/it]
all        64        1105      0.678      0.661
0.691      0.469

```

```

Epoch      GPU_mem    box_loss    cls_loss    dfl_loss    Instances
Size
48/50      12.3G      0.6396      0.5982      0.8849      200

```

```

640: 100% 15/15 [00:13<00:00, 1.15it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 1.01it/s]
      all         64      1105       0.73       0.638
0.702      0.483

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      49/50       12.3G      0.6426     0.5856     0.8817       193
640: 100% 15/15 [00:14<00:00, 1.02it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:00<00:00, 1.13it/s]
      all         64      1105       0.682       0.666
0.687      0.461

```

```

      Epoch      GPU_mem    box_loss    cls_loss    dfl_loss  Instances
Size
      50/50       12.3G      0.6285     0.5793     0.8787       303
640: 100% 15/15 [00:13<00:00, 1.10it/s]
      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:01<00:00, 1.32s/it]
      all         64      1105       0.735       0.639
0.697      0.476

```

50 epochs completed in 0.220 hours.

Optimizer stripped from runs/detect/train3/weights/last.pt, 22.5MB

Optimizer stripped from runs/detect/train3/weights/best.pt, 22.5MB

Validating runs/detect/train3/weights/best.pt...

Ultralytics 8.3.105 Python-3.11.11 torch-2.6.0+cu124 CUDA:0 (Tesla T4, 15095MiB)

Model summary (fused): 72 layers, 11,130,228 parameters, 0 gradients, 28.5 GFLOPs

```

      Class      Images Instances      Box(P          R
mAP50  mAP50-95): 100% 1/1 [00:01<00:00, 1.31s/it]
      all         64      1105       0.732       0.644
0.713      0.492
      class_1       12         26       0.79       0.423
0.649      0.448
      class_3         7         29       0.712       0.69
0.617      0.368
      class_4       11         91       0.363       0.418
0.311      0.217
      class_5       52         56       0.924       0.839
0.899      0.43
      class_6       38         68       0.878       0.632
0.791      0.453
      class_7       14         35       0.8       0.8
0.828      0.741
      class_8       49        210       0.734       0.774

```

0.826	0.477				
	class_9	16	22	0.72	0.636
0.712	0.644				
	class_10	57	561	0.77	0.8
0.857	0.668				
	class_11	3	7	0.627	0.429
0.639	0.477				

Speed: 0.2ms preprocess, 5.3ms inference, 0.0ms loss, 2.4ms
postprocess per image

Results saved to runs/detect/train3

□ Learn more at <https://docs.ultralytics.com/modes/train>